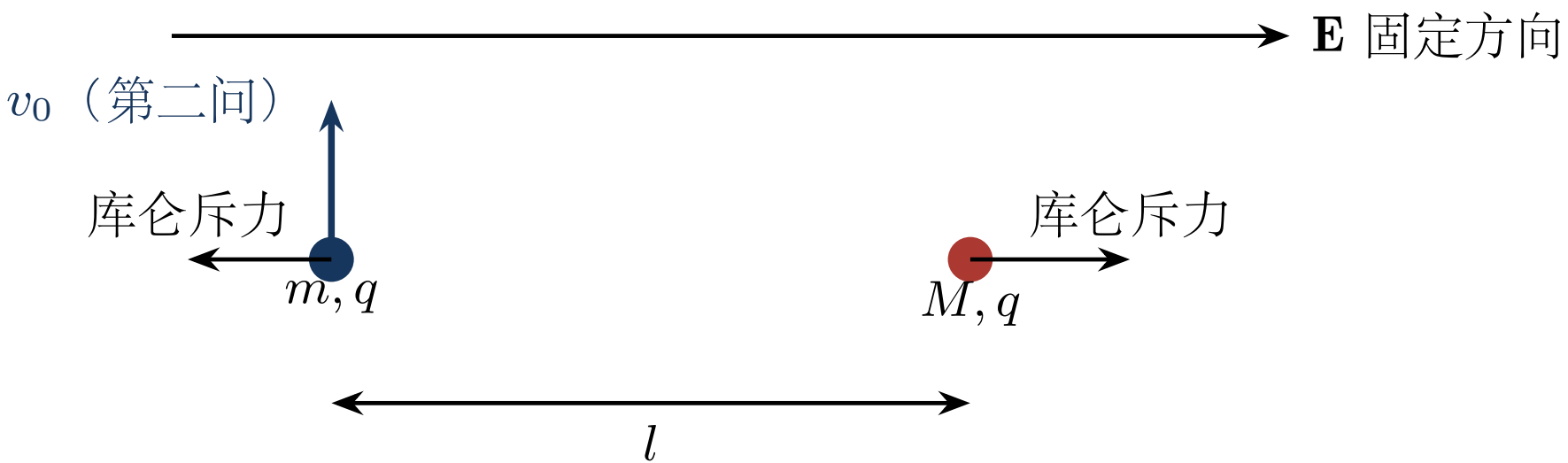

相对坐标 $\mathbf{r} = \mathbf{r}_M - \mathbf{r}_m$

初相对速度为 $(0, -v_0)$